



SCHOOL *of* DATA SCIENCE



Troubleshooting

DS 6015: Data Science Capstone
School of Data Science
University of Virginia

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Troubleshooting

Review of common issues encountered by earlier teams

Potential solutions are proposed

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Not Enough Time

Q: What if there isn't enough time to complete capstone?

- > Capstone projects are generally scoped to fit
- > Projects kick off the term before to allow for onboarding and discovery.
Teams that use this time wisely tend to do better.
- > If you are waiting on feedback, data, etc., stay productive.
What else can be done?
- > Share the work across students

Unclear Problem Definition

Each project should have a problem statement, but clarity may vary.

Q: What if the sponsor doesn't clearly define the problem?

You might propose ideas at intersection of what's achievable & interesting

Lack of Data

Each project should have data, but coverage and quality may vary

Q: What if the data won't support the objective?

You might propose ideas on what is achievable

- Solve for subset of population?
- Pivot the question?
- If labeled dataset is small, you might train on labeled data, predict on unlabeled, and review cases with high/low probability.
- Can more data be collected? Can supplemental data be created?

Making Meetings more Productive

Each sponsor should meet w students as needed.

Q: What if the meetings aren't productive enough?

You might try:

- Forming an agenda / agreeing on it beforehand
- Sharing results beforehand to discuss during meeting
- Decide Next Steps in the meeting
- Call out decision points in meeting
- Ask for feedback

High Workload

The sponsor is made aware that capstone is one of several things students are working on. At the same time, capstone is an important component

Q: What if the sponsor demands cannot be met?

First, the students should discuss and decide if things truly are too demanding, and assess if you are managing your time well.

You can discuss with your faculty mentor and Capstone Director

You can gently remind the sponsor of the workload

You can suggest what might be more reasonable and get buy in

Ideally, the project brings together skills you've learned in the program.

Q: What if I don't have the skills to solve the problem?

- > In some cases, you will need to learn new things
- > Your sponsor / mentor might be able to suggest resources

Ideally, the project brings together skills you've learned in the program.

Q: What if I don't have the skills to solve the problem?

- > In some cases, you will need to learn new things
- > Your sponsor / mentor might be able to suggest resources
- > Keep a list of open questions, skills needed
 - Break these down into manageable tasks (e.g., learn n-gram models)
- > You might distribute the work across teammates and share learnings

Sparse Feedback

Your mentor and sponsor should provide feedback on your work

Q: What if feedback isn't arriving often enough?

You can try:

- > Checking if there is a better medium for feedback (email, meetings, .
- > Asking if there is a better way to share results
 - Are questions clear enough?
 - Are things summarized?
(e.g., assemble separate questions into single slide)

Sparse Feedback

Your mentor and sponsor should provide feedback on your work

Q: What if feedback isn't arriving often enough?

You can try:

- > Understanding if there are busier/quieter periods
- > Working on aspects not requiring the feedback

Model Doesn't Generalize

You've built a model on training data
The performance on new data is worse

Q: Why does my model perform much worse on new data?

This is usually a symptom of overfitting. Can consider:

- k-fold cross validation (for ML models)
- Dropout layers (for deep neural networks)
- More suitable data (larger dataset, more recent data)
- Looking at the distribution of training data, new data.
Is there drift in distribution? If so, training set isn't sufficient.

Unstable Model

You've built a model

When you change hyperparameters slightly, the results change drastically

Q: Why is my model so unstable?

While we tune models for better performance,
a drastic change in output is not desirable

> The model may be too complex

Can try simpler model and examine sensitivity

> Did you cross validate the model? This is a good practice when feasible

Sponsor Unsure how to use Deliverables

Q: What if the sponsor is unsure how to use the deliverables?

- > Share work early and often. Avoid dropping everything on them at the end.
- > Leave time at the end to walk sponsor through deliverables
- > Have a meeting where sponsor team runs the code
- > Clearly document the work / code. A README file can be very effective. It can include objective, main functionality.

Final Thoughts

You have a support system to help you through capstones including:

- teammates
- sponsor
- faculty mentor
- Program Director (Adam Tashman)

Please lean on this system for help!